

Lentil (Masoor) Farming Guide for Pakistan

A complete crop guide for farmers, students, and agriculture advisors

1. Introduction

Lentil, known locally as Masoor, is one of the most popular pulse (legume) crops in Pakistan, forming a staple part of the national diet in the form of daal. It is a cool-season crop grown mainly during the winter months, valued for its high protein content, affordability, and importance in vegetarian and mixed diets across the country. Lentil is grown both as a sole crop and, in some areas, intercropped with other winter crops. Like other legumes, lentil improves soil fertility by fixing atmospheric nitrogen, making it a valuable crop in rotation with cereals such as wheat. Despite its importance, Pakistan still imports a significant portion of its lentil requirement, which highlights the opportunity for expanding domestic production.

2. Scientific Name

The scientific name of lentil is *Lens culinaris*, belonging to the family Fabaceae (legume family). It is commonly known as Masoor in Urdu and other local languages across Pakistan.

3. Importance in Pakistan

Lentil (Masoor) is an essential source of dietary protein for millions of people in Pakistan, especially in households that rely on pulses as an affordable alternative to meat. Masoor daal is a common part of daily meals across all provinces, valued for its taste, easy cooking, and nutritional benefits, including protein, iron, and fiber. Beyond its role in the diet, lentil cultivation benefits soil health through nitrogen fixation, improving fertility for subsequent cereal crops. Lentil also requires relatively low input costs compared to many other crops, making it an accessible option for smallholder farmers, particularly in rain-fed and semi-arid areas where water availability is limited during the winter cropping season.

4. Major Producing Provinces and Districts

Lentil cultivation in Pakistan is concentrated in specific regions suited to its cool-season growth habit. - **Punjab:** Major lentil-growing districts include Chakwal, Jhelum, Attock, Mianwali, and parts of the Pothohar region, mostly under rain-fed conditions. - **Sindh:** Grown on a smaller scale in some districts under irrigated conditions. - **Khyber Pakhtunkhwa:** Districts such as Kohat, Karak, and Bannu grow lentil, mainly as a rain-fed winter crop. - **Balochistan:** Grown in scattered areas with suitable winter rainfall, including parts of Quetta and Pishin. Punjab's Pothohar region remains the leading area for lentil production in the country.

5. Climate Requirements

Lentil is a cool-season crop that grows best in temperatures ranging from 18°C to 30°C, with cooler temperatures around 15°C to 20°C being ideal during the vegetative growth stage. It requires a relatively dry climate during flowering and pod maturity, as excess moisture or humidity at these stages can encourage disease and reduce seed quality. Lentil is sensitive to frost during flowering, though mature plants can tolerate light frost better than during early flowering stages. It grows well with modest rainfall of about 300 to 400 millimeters during the growing season, mostly relying on winter rains in Pakistan's rain-fed lentil-growing areas.

6. Soil Requirements

Lentil grows best in well-drained loamy to sandy loam soils, though it can also perform reasonably well in clay loam soils as long as drainage is adequate. It is sensitive to waterlogging, which can severely damage roots and reduce yield, so fields with poor drainage should be avoided or improved before sowing. The ideal soil pH range for lentil is between 6.0 and 8.0, and it tolerates mildly alkaline soils common in parts of Punjab and Sindh reasonably well. Fertile soils with good organic matter content support better nodulation and yield.

7. Land Preparation

Good land preparation is important for lentil to ensure proper seed-to-soil contact and root development. The field should be ploughed two to three times to achieve a fine, well-pulverized seedbed, followed by planking to level the surface and conserve soil moisture, which is especially important in rain-fed cultivation. In areas prone to waterlogging, care should be taken to ensure proper field drainage, since lentil roots are highly sensitive to excess water. Land preparation should be completed in time for sowing at the start of the cool season to maximize the use of available soil moisture.

8. Recommended Varieties (Pakistan)

Several improved lentil varieties have been released for cultivation across Pakistan. - **Masoor-2006:** A widely grown, high-yielding variety recommended for Punjab. - **Punjab Masoor-2009:** Known for good disease resistance and yield stability. - **NIAB Masoor-2002:** Developed by the Nuclear Institute for Agriculture and Biology (NIAB), Faisalabad, popular among Punjab farmers. - **Markaz Masoor-2009:** Suited to rain-fed conditions of the Pothohar plateau. Farmers are encouraged to source certified seed of these recommended varieties from provincial seed corporations to ensure good germination and disease resistance.

9. Seed Rate

The recommended seed rate for lentil depends on seed size and sowing method. - For drilling in rows: 12 to 15 kilograms per acre. - For broadcasting: 15 to 18 kilograms per acre, since broadcasting typically results in less uniform seed placement and slightly lower germination efficiency. Using good quality seed with high germination percentage helps achieve the desired plant population and reduces the need for excessive seed rates.

10. Seed Treatment

Before sowing, lentil seed should be treated with a fungicide such as Thiram or Carbendazim to protect against seed-borne diseases and improve seedling emergence. As a legume, lentil seed also benefits significantly from treatment with Rhizobium bacterial inoculant, which enhances nitrogen fixation, especially in fields where lentil or related pulses have not been grown recently. Seed should be treated gently to avoid damaging the seed coat, and inoculated seed should be sown promptly, ideally on the same day, for best results.

11. Sowing Time (Province-wise)

- **Punjab:** Lentil is typically sown from mid-October to mid-November in the Pothohar region, taking advantage of early winter moisture.
- **Sindh:** Sowing under irrigated conditions generally takes place from October to November.
- **Khyber Pakhtunkhwa:** Sowing is usually done from October to early November, depending on local temperature patterns.
- **Balochistan:** Sowing typically occurs in October, aligned with the onset of the cooler season. Timely sowing is important to ensure the crop establishes well before the coldest part of winter and completes flowering before the onset of hot weather in spring.

12. Sowing Methods

Lentil can be sown using different methods depending on farm resources. - **Drilling with seed drill:** Preferred method, sowing seed in rows with spacing of about 25 to 30 centimeters, allowing for easier weeding and better plant stand uniformity. - **Broadcasting:** Still common among smallholder farmers, particularly in rain-fed areas, though it results in less uniform spacing. - **Sowing behind the plough:** A traditional method where seed is dropped into furrows created by a plough, still practiced in some rain-fed regions. Row sowing with a seed drill generally produces better yield due to more efficient use of moisture and nutrients, as well as easier weed control.

13. Fertilizer Recommendations (NPK)

Lentil requires relatively modest fertilizer inputs due to its nitrogen-fixing ability, but responds well to phosphorus. - **Nitrogen (N):** 10 to 15 kilograms per acre, applied as a starter dose at sowing to support early growth before nitrogen fixation becomes fully active. - **Phosphorus (P2O5):** 40 to 50 kilograms per acre, applied fully at sowing time, since phosphorus is critical for root development and nodulation in legumes. - **Potassium (K2O):** 10 to 15 kilograms per acre, particularly on soils known to be low in potassium. Proper Rhizobium inoculation combined with adequate phosphorus application significantly improves nodulation, nitrogen fixation, and overall yield.

14. Irrigation Schedule

Under irrigated conditions, lentil requires only two to three irrigations during its growing season, since it is relatively drought tolerant once established. The first irrigation, if needed, is usually given at the pre-flowering stage, followed by a second irrigation during pod formation, which is the most critical stage for water availability. Excess irrigation, especially near maturity, should be avoided as it can promote disease and reduce seed quality. In rain-fed areas, no supplemental irrigation is applied, and the crop depends entirely on winter rainfall.

15. Weed Management

Weeds can significantly reduce lentil yield if not controlled, especially during the first 30 to 45 days after sowing when the crop grows slowly. Effective weed management includes one or two hand weeding or hoeings during this critical early period. Pre-emergence herbicides can also be used on larger farms, applied according to label instructions and local recommendations, followed by manual weeding if needed later in the season. Since lentil plants are relatively short and have limited ability to compete with tall weeds, timely weed control is especially important for good yield.

16. Insect Pests

Several insect pests can affect lentil crops in Pakistan. - **Aphids:** Suck sap from leaves and stems, weakening plants and potentially transmitting viral diseases. - **Pod borer:** Larvae feed on developing pods, directly reducing seed yield and quality. - **Cutworms:** Occasionally damage young seedlings by cutting stems near ground level. - **Whitefly:** Can build up in warmer conditions, weakening plants and spreading viral diseases. Regular monitoring and timely application of recommended insecticides, only when pest levels justify treatment, help protect yield while minimizing chemical costs and environmental impact.

17. Diseases

Lentil crops in Pakistan can be affected by several important diseases. - **Rust:** Causes reddish-brown pustules on leaves and stems, reducing photosynthetic capacity and yield. - **Wilt (Fusarium wilt):** A soil-borne fungal disease causing sudden wilting and death of plants, especially in warm, dry conditions. - **Blight (Ascochyta blight):** Causes lesions on leaves, stems, and pods, particularly under humid conditions. - **Root rot:** Affects seedlings and mature plants in poorly drained or waterlogged soils. Using resistant varieties, proper crop rotation with non-legume crops, and avoiding waterlogged conditions are key strategies for managing these diseases.

18. Deficiency Symptoms

Nutrient deficiencies in lentil present recognizable symptoms. - **Nitrogen deficiency:** General yellowing of older leaves, though typically less severe due to nitrogen fixation once nodulation is established. - **Phosphorus deficiency:** Dark green to purplish leaf color, along with poor root and nodule development and delayed flowering. - **Potassium deficiency:** Yellowing and browning along leaf margins, with weaker stems. - **Iron deficiency:** Common in alkaline soils, causing yellowing between veins on younger leaves. - **Sulfur deficiency:** Can cause general pale yellowing of young leaves, sometimes confused with nitrogen deficiency. Correcting deficiencies through balanced fertilization and proper Rhizobium inoculation supports healthy growth and pod development.

19. Harvesting

Lentil is ready for harvest when the lower pods turn brown and the plant leaves begin to dry and shed, usually 120 to 150 days after sowing depending on variety and sowing date. Harvesting is typically done by cutting plants close to the ground using a sickle, since lentil pods can shatter easily if left too long in the field, leading to seed loss. After cutting, plants are left to dry in the field or on a threshing floor for a few days before threshing, which can be done manually by beating or using a mechanical thresher. Careful timing of harvest, avoiding both premature and delayed harvesting, helps

minimize shattering losses and maintain seed quality.

20. Storage

After threshing, lentil seed should be cleaned and dried thoroughly to a moisture content of around 10 to 12 percent before storage, to prevent insect infestation and fungal growth. Seed should be stored in clean, dry, well-ventilated bags or containers, kept off the ground, and protected from moisture and pests such as pulse beetles, which are a common storage pest for lentil. Regular inspection during storage and, if necessary, use of approved grain protectants help maintain seed quality and germination for future planting or consumption.

21. Expected Yield in Pakistan

Under rain-fed conditions in Punjab's Pothohar region, average lentil yield ranges from about 6 to 10 maunds per acre (225 to 375 kilograms per acre), depending on rainfall and management practices. Under good irrigated conditions with improved varieties and proper fertilization, yields can reach 12 to 15 maunds per acre (450 to 560 kilograms per acre) or slightly higher. Actual yield depends significantly on rainfall timing during the growing season, variety selection, and effective disease and pest management.

22. Government Recommendations

Provincial agriculture research institutes, including the Barani Agricultural Research Institute (BARI) Chakwal and the Ayub Agricultural Research Institute (AARI) Faisalabad, regularly issue recommendations for lentil cultivation. These typically include guidance on using certified seed of recommended varieties, applying Rhizobium inoculant for better nitrogen fixation, and following recommended sowing windows to maximize the use of winter rainfall. Farmers are also encouraged to include lentil in crop rotation with wheat and other cereals to improve overall soil fertility, and to seek advice from local agriculture extension offices regarding the latest variety releases and pest management recommendations.

23. Modern Farming Practices

Farmers growing lentil in Pakistan are increasingly adopting improved practices such as using seed drills for uniform row sowing instead of traditional broadcasting, applying Rhizobium inoculants to boost yield without increasing fertilizer costs, and using combine harvesters on larger farms to reduce harvest losses from pod shattering. Improved seed cleaning and storage practices are also being promoted to reduce losses from pulse beetle infestation and to maintain better seed quality for both consumption and future planting.

24. Precision Agriculture

Precision agriculture techniques are gradually being introduced for lentil cultivation in Pakistan through research and extension initiatives. These include soil testing to guide precise phosphorus and potassium application rather than blanket fertilizer recommendations, and moisture monitoring in irrigated areas to optimize the timing of the limited irrigations lentil requires. Remote sensing and satellite-based crop monitoring are also being explored by research institutions to estimate lentil area and yield across rain-fed districts, helping guide better planning and support for farmers in the future.

25. Sustainable Farming

Lentil is inherently supportive of sustainable farming due to its nitrogen-fixing ability, which reduces the need for synthetic nitrogen fertilizer and improves soil fertility for following crops such as wheat. Sustainable practices encouraged in lentil cultivation include crop rotation with cereals to break pest and disease cycles, minimum tillage to conserve soil moisture in rain-fed areas, and integrated pest and disease management to reduce reliance on chemical inputs. Because lentil requires relatively little irrigation, it also supports more efficient use of scarce water resources, particularly important in Pakistan's water-stressed agricultural regions.

26. Summary Note

This guide reflects lentil (Masoor) farming practices specific to Pakistan, with particular focus on the Pothohar region of Punjab and other growing areas in Sindh, Khyber Pakhtunkhwa, and Balochistan. Farmers should confirm current variety recommendations, seed availability, and sowing schedules with their local agriculture extension office, as these are updated periodically based on ongoing research.